PERMEABILITY USING MDCK-LE VERSION 2 MONOLAYER

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LIST OF ABBREVIATIONS

ABBREVIATION	DESCRIPTION
%	Percentage
°C	Degree Celsius
µM	Micromolar
µm	Micrometer
A to B	Apical to basolateral
ACN	Acetonitrile
B to A	Basolateral to apical
$CaCl_2$	Calcium chloride
Caco-2	Colon adenocarcinoma derived cells
cm^2	Centimeter squared
Co	Initial dosing concentration
COA	Certificate of analysis
DMEM	Dulbecco's modified eagle medium
DMPK	Drug metabolism pharmacokinetics
DMSO	Dimethyl sulfoxide
EDTA	Ethylene di amine tetra acetic acid
FBS	Fetal bovine serum
g	Gram
H_2O	Dihydrogen monoxide
HBSS	Hank's buffered salt solution
HCL	Hydrochloric acid
HEPES	4-(2-Hydroxyethyl)piperazine-1-ethanesulfonic acid, N-(2-hydroxyethyl) piperazine-N'-(2-ethanesulfonic acid)
HOD	Head of the department
IPM	Intellectual property management
KCl	Potassium chloride
KH_2PO_4	Potassium dihydrogen phosphate
L	Litre
LAF	Laminar air flow
LC-MS/MS	Liquid chromatography coupled with tandem mass spectrometry
LNB	Lab note book
MDCK-LE v2	Madin-Darby canine kidney low efflux version 2 cell line
MDCK-WT	Madin-Darby canine kidney wild type cell line
mg	Milligram
$MgCl_2$	Magnesium chloride
$MgSO_4$	Magnesium sulfate
min	Minutes

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ABBREVIATION	DESCRIPTION
mL	Milliliter
mM	Millimolar
mm	Millimeter
MRP	Multi drug resistance protein
MS	Microsoft
N	Normality
NA	Not applicable
Na_2HPO_4	Disodium hydrogen phosphate
NaCl	Sodium chloride
$NaHCO_3$	Sodium bicarbonate
NaOH	Sodium hydroxide
PBS	Phosphate buffered saline
PD	Pharmaceutical development
p-gp	P-glycoprotein
pH	Negative logarithm to base 10 of the positive hydrogen ion molar concentration
PPE	Personal protective equipment
RFU	Relative fluorescence unit
rpm	Revolutions per minute
RT	Room temperature
S. No.	Serial number
US	United states
MDR1	Multidrug Resistance Protein 1 or P-gp

1. OBJECTIVE

The objective of this experiment is to determine the apparent permeability of test compound(s) using MDCK-LE version 2 monolayer.

2. SCOPE

The procedure described in this protocol will be used for testing the apparent permeability of test compound(s).

3. RESPONSIBILITY

All personnel performing this study are responsible to follow the process outlined in this protocol, at Aurigene Pharmaceutical Services Ltd, India.

4. INTRODUCTION

Assessment of the intestinal permeability of compounds intended for oral administration plays an important role in selecting candidates for drug development. Ranking a series of lead compounds in order of absorption potential facilitates compound selection and optimization. A currently accepted method for investigation of the absorption potential of compounds within a series is by comparison of the apparent permeability through Caco-2 (Human colon carcinoma) or MDCK (Madin Darby Canine Kidney) monolayer cultures These absorption models are also useful for understanding any absorption issues associated with compounds further advanced in development, including those involved with active transport mechanisms. MDCK cells differentiate into columnar epithelium and form tight junctions when cultured on semi-permeable membranes.

The primary concern of using MDCK cells is the presence of endogenous transporters of non-human origin. The canine P-glycoprotein (P-gp) can interfere with permeability and transporter studies, leading to less reliable data. MDCK-LE version 2 cell line developed by knockout the canine MDR1 gene from a subpopulation of MDCK-WT cells having low-efflux which has significantly lower canine P-gp expression and shows less functional efflux activity than MDCK-WT based on efflux ratios. MDCK-LE version 2 cells offer clear advantages over the MDCK-WT by providing less efflux transporter background signals and minimizing interference from canine P-gp. Therefore MDCK-LE version 2 permeability assay measures the passive permeability of compounds across a monolayer and is high throughput with good precision, reproducibility, robustness, cost-effectiveness.

5. SAFETY AND PRECAUTIONS

During the conduct of the study, the study personnel should take necessary safety precautions and use personal protective equipment (PPE) such as facemasks, gloves, lab coat, goggles, shoes etc.

MDCK cell line is classified under Bio-Safety Level 1 based on the US Public Health Service Guidelines. It is the responsibility of the user to ensure that their facilities comply with the recommended biosafety regulations.

DMSO:

- Keep away from sources of ignition.
- Do not breathe vapour or mist.
- Avoid contact with skin, eyes and clothing.

6. EQUIPMENT AND CHEMICALS

6.1 EQUIPMENT

Equipment	Make
96 well deep well plates (1.2, 2 mL)	Axygen Scientific

Equipment	**Make**
Analytical balance	Sartorius/ Denver
Biohazard safety cabinet class II	ESCO
Cell culture flasks (T-25, T-75 and T-150 cm^2)	Corning, Eppendorf
Centrifuge with plate/ tube rotors	Eppendorf/ Thermo/ Kubota
Corning HTS Transwell-96 System plates	Corning; CLS 3392
Cryo vials	Nalgene
Cyclomixer	IKA® Vortex 3
Deep freezer (-80°C)	Thermo Scientific/ Sanyo
Glass bottles (100, 250, 500 and 1000 mL)	Schott DURAN
Glass cover slips	Blue star
Haemocytometer	Rohem
Incubator	Heat Control Instruments & Services
Inverted microscope	Nikon
LC-MS/MS (API-5500, API-4000, API-3200)	AB Sciex
Micro centrifuge tubes	Tarsons
Micro-titre plate shaker	IKA® MTS
Multi-channel pipettes(P-1200)	Eppendorf
pH meter	Horiba scientific
Plate shaker	Heidolph
Refrigerator	LG
Single channel pipettes (p 10, 100, 200 1000)	Eppendorf
Software	Microsoft Excel
Sonicator	Shalom instruments
Step 1 cooler	Thermo Scientific
Sterile bottle-top filter	Nalgene
Sterile conical falcon tubes (15 and 50 mL)	Tarsons
Sterile disposable pipettes (2, 5, 10 and 50 mL)	Eppendorf
Steripette	Thermo Scientific
Water bath	Shalom instruments
Water-jacketed CO_2 incubator	Eppendorf

Note:

- Equivalent equipment/ instruments could be used based on the availability.

6.2 CHEMICALS

Chemicals	Make; Catalogue No.
0.05 % Trypsin, EDTA, 1x	Gibco; 25300-054
Acetonitrile	J.T. Baker; 9829-03
BafilomycinA1	Sigma; B1793
Bestatin	Sigma; B8385
BSA	Sigma; A9647
DMEM	Sigma; D5796
DMSO	Sigma Aldrich; D8418
DPBS 1X	Gibco; 14190144
Ethanol	Changshu hongsheng fine chemicals Co, Ltd
Fetal bovine serum	Gibco; 10437028
Glyburide	Sigma Aldrich; G2539
HBSS 1X	Gibco; 14025092
HEPES 1M	Gibco; 15630080
Penicillin-streptomycin solution 100 X	Invitrogen; 15140122
Puromycin 10 mg/ml	Gibco; A11138-03
Quinidine	Sigma; Q0750
Sodium hydroxide	Himedia; GRM1183
Trypan blue	Sigma; T8154

Note:

- Complete list of chemicals used, and their details should be provided in the LNB.
- Equivalent grade chemicals/ reagents from alternate sources can be used on need basis.

6.3 PREPARATION OF SOLUTIONS AND STORAGE CONDITIONS

S. No.	Composition		Storage condition
1.	Preparation of culture media for growing MDCK-LE version 2 cells		2-8°C
	DMEM liquid media	1000 mL	
	Penicillin-streptomycin solutions (100 X)	11 mL	

S. No.	Composition	Storage condition
	FBS 55 mL Puromycin (10 mg/ml) 280 µL Mix contents well and label the bottle with the name of media, components, and the date of preparation. The media should be used within one month after which it has to be discarded.	
2.	Preparation of Bafilomycin solution Prepare a 100 µM stock solution in Ethanol	-20°C
3	Preparation of bestatin solution Prepare a 10 mM stock solution in DMSO	
4.	Preparation of Assay buffer(5%BSA) Prepare the assay buffer by mixing with 5% BSA, 10Mm HEPES and 0.1 µM Bafilomycin	2-8°C
5.	Preparation of Assay buffer (0.02% BSA) Prepare the assay buffer by mixing HBSS with 0.02% BSA, 10 mM HEPES and 0.1µM Bafilomycin	Prepare fresh
6.	Preparation of compound buffer Prepare compound buffer by adding 10 µM bestatin to assay buffer (0.02% BSA). This solution is used to prepare the working solutions of test & reference compounds.	Prepare fresh
7.	Preparation of working solutions of test/ reference compound(s) Prepare 10 mM stock solutions in DMSO and dilute it further (0.4% DMSO) to get 10 µM in compound buffer containing Bestatin. If compounds are screened as a pool, a cocktail of 3 compounds each 10 µM, should be prepared. Centrifuge these working stock solutions for 10 min at 4000 rpm at RT and collect the supernatant. Use this supernatant for the assay incubations. Any compounds which show precipitation should not be included in the assay.	Prepare fresh
8.	Preparation of calibration standards of test/ reference compound(s) Prepare 10 mM stock solutions in DMSO and dilute it to get 750 µM in DMSO containing Bestatin and dilute it further in DMSO to obtain the following working stock concentrations 750, 187.5, 46.88, 11.72, 2.93, 1.46 and 0.73 µM. Dilute the working solution in HBSS buffer (5% BSA) to obtain final concentrations of 3000, 750, 187.5, 46.88, 11.72, 5.86 and 2.93 nM. If compounds are screened as a pool, a cocktail of calibration standards of 3 compounds should be prepared (0.4% DMSO). Centrifuge these calibration stock solutions for 10 min at 4000 rpm at RT and collect the supernatant. Use this supernatant for the further processing.	Prepare fresh

S. No.	Composition	Storage condition
9.	Preparation of stop solution Prepare 1 mg/mL stock solution of the internal standard such as Glyburide and dilute with acetonitrile to obtain appropriate concentration (125 ng/ mL)	2-8°C

7. PROCEDURE

7.1 GROWING MDCK-LE VERSION 2 CELLS IN CULTURE

Remove a vial of MDCK-LE version 2 cells from the liquid nitrogen cryo tank and wiped with 70% isopropanol.

The vial is quickly thawed at 37°C by immersing in water bath.

Take the vial to the biosafety cabinet and transfer the thawed content to a sterile 50 mL tube containing DMEM media (without puromycin). After brief mixing, tube is centrifuged at 500 rpm for 5 min at 20- 25°C.

Supernatant is decanted and cell pellet is re-suspended in 2 mL of DMEM media (without puromycin) and transfer the contents into a T-75 flask containing 13 mL of pre-warmed DMEM media (with puromycin). Place the flask in the CO_2 incubator.

Change the media on alternate days. Upon reaching 70- 80% confluency cells are revived. It is ideal to passage the cells at least once after reviving before seeding.

7.2 TRYPSINIZATION OF CELLS FOR FLASK

Take the flask with 70-80% confluent MDCK-LE version 2 cells inside the biosafety hood.

Aspirate the media and discard it.

Add 10 mL of DPBS that is pre-warmed to 37°C. Rinse the cells by gently tilting the flask. Aspirate the wash buffer and repeat the step twice.

Add 1 mL of 0.05% trypsin-EDTA solution that is pre-warmed to 37°C. Rinse the cells by gently tilting the flask. Aspirate the trypsin and discard it.

Add 3 mL of 0.05% trypsin-EDTA solution. Cap the flask and transfer it to the CO_2 incubator. Incubate for about 5-10 min.

Bring the flask back to the biosafety hood. Gently tilt the flask several times to dislodge the cells.

Check the flask using a microscope. Once the cells have completely detached, add 30 mL of DMEM media (without puromycin) to the flask. Using a sterile pipette, disperse the cells by gentle but thorough pipetting of the trypsin solution in the flask. This helps to disperse any cell clumps and provides a homogeneous cell suspension.

Transfer the contents of the flask to a sterile 50 mL sterile centrifuge tube. Centrifuge the tube at 500 rpm for 5 min at 20- 25°C.

Take the tube back to the biosafety hood and wipe the outer surface with 70% isopropanol. Aspirate the supernatant without disturbing the cell pellet.

Add 5 mL of fresh DMEM media (without puromycin) to the tube and re-suspend the pellet by gentle but thorough pipetting.

Dilute an aliquot of the cells and count the viability of cells by trypan blue staining.

Transfer the appropriate volume of cell suspension to seed a new flask containing fresh DMEM medium (with puromycin) and place the flask into the cell culture incubator.

7.3 TRYPAN BLUE STAINING FOR VIABLE CELL COUNT

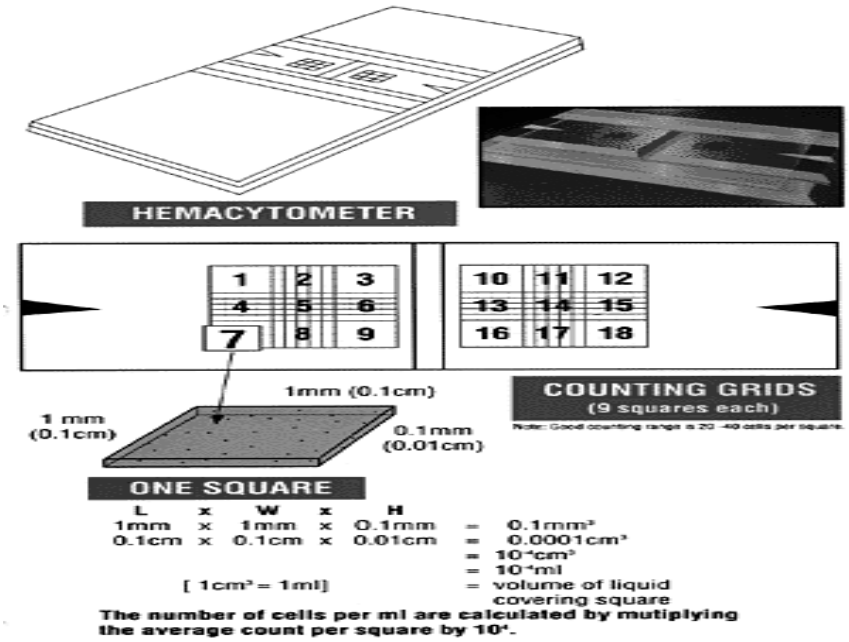
Load both sides of the haemocytometer with the cell suspension covering both counting grids.

Count the number of cells in 4 out of 9 squares from each counting grid (count a total of 8 squares). Divide the total number of cells counted from 8 squares by 8 to get the average number of cells per square.

Multiply the average count per square by 10^4 (multiply with dilution factor if any dilution) to get the number of cells per mL of the original cell suspension.

Number of cells per mL (million/ mL) = average count per square × dilution factor × 10^4.

To determine the total number of cells available, multiply the number of cells per mL by the total mL of cell suspension.



7.4 PREPARATION OF FILTER PLATE FOR CELL SEEDING

Warm all the required solutions to 37°C in a water bath/ incubator.

Ahead of seeding HTS Transwell®-96 Tissue Culture Systems (Corning®, 0.143 cm^2 surface area) are preincubated with DMEM media.

Add 75 µL of DMEM media (without puromycin) to the wells in the apical 96 well permeable support plate and DMEM media (without puromycin) 30 mL to the basal reservoir plate/ 235 µL to receiver plate.

Cover the plate and place in the CO_2 incubator 37°C for at least 30 min until trypsinized cells are ready.

7.5 SEEDING OF CELLS

Aspirate the medium from the lower and upper compartments of the filter plate

Dilute the cell suspension with pre-warmed DMEM media (without puromycin) to get 50,000 cells/mL.

Dispense 75 µL of cell suspension (3,750 cells/ well) to the wells in the apical 96 well plate and DMEM media (without puromycin) 30 mL to the basal reservoir plate/ 235 µL to receiver plate.

Plates are transferred to incubator at 5% CO_2 and 37°C and incubate for 5 days.

Thereafter, cells are fed with fresh media (without puromycin) at 48-72 hrs.

On the day of the assay, monolayer is washed thrice with prewarmed 1 X HBSS.

7.6 SCREENING OF TEST COMPOUND(S) FOR PERMEABILITY

Wash the cells 3 times with HBSS.

10 mM compound stock solutions are prepared in 100% DMSO.

Working stock solutions are prepared as given in section-6.3

Test compounds are tested only in A-B direction by adding 75 µL working stock solutions to the apical 96 well plate and 235 µL assay buffer (5% BSA) to the corresponding wells of basal receiver plate.

The reference compound quinidine is tested in the bidirectional assay. For A to B direction, 75 µL of the working stock solution is dispensed in the apical wells and 235 µL of assay buffer (5% BSA) to the basal receiver wells. For the B to A transport, dispense 235 µL of the working stock solution in basal wells and 75 µL of assay buffer (5% BSA) in the corresponding apical wells.

Test compounds are pooled together in a cassette of 3 compounds with 10 µM bestatin.

Test compound and reference compounds incubations were in tested in triplicates.

Incubate the sandwich for 120 min in incubator at 5% CO_2 and 37°C.

7.7 SAMPLE PROCESSING

After incubation, transfer 60 µL from receiver and donor compartments to collection plate.

Transfer 20 µL of all samples in to 96 well V- bottom dispense plates and normalise the matrix by adding 20 µl assay buffer (5.0% BSA) for donor, Co and calibration curve samples. Similarly add 20 µl assay buffer (0.02% BSA) for acceptor samples

Add 160µl stop solution (refer section-6.3)

Vortex the samples. Centrifuge the plates at 4000 rpm, 4°C for 30 min.

Transfer 150 µL of supernatant and add 100 µL HPLC water seal the plate.

Analyse the compound concentration in all samples using respective calibration standards by LC-MS/MS.

Note:

- Any changes in the procedure should be captured in LNB.
- Replicates and test concentration can be only modified with due approval of the group leader.
- Choice of buffers, pH and temperature can be modified as requested by the end user.

8. ASSAY CONTROL

Quinidine and Bestatin should be included as assay control.

Study would be qualified based on the acceptance criteria given in the data template.

Note:

- Any changes to the assay controls would be made only with the consent of the sponsor and should be described in the LNB & report ((If any).

9. DATA RECORDING

The data should be tabulated in MS excel based MDCK-LE version 2 permeability assay template and the final data should be pasted in the LNB. Raw data should be recorded in the LC-MS/MS server with the study number and the data path should be provided in the excel file and LNB.

10. DATA ANALYSIS

Calculate the P_{app} and recovery using the given formula:

$$\text{Permeability (cm/s): } P_{app} = \frac{dQ}{dt} \times \frac{V}{C_0} \times \frac{1}{A}$$

$$\text{Recovery \%} = 100 \times \frac{(\text{donor volume x Cend,d}) + (\text{acceptorvolume x Cend,a})}{\text{donar volume x C HBSS}}$$

Where,

dQ/dt = Permeability rate in μM/sec

C_0 = Initial concentration in μM

A = Membrane surface area (0.143 cm^2)

V = Volume of receiver chamber

$C_{end, a}$ = concentration in acceptor well at end of incubation

$C_{end, d}$ = concentration in donor well at end of incubation

C_{HBSS} = loading concentration measured in HBSS

11. REPORTING

Data should be reported in an MS excel format. Based on the need, MS word-based reports may also be prepared.

If reference compound(s) does not meet the specified acceptance criteria, the complete batch should be rejected.

Membrane Integrity Control: **Bestatin mean Papp < 2.0** (10^{-6} cms^{-1})
Reference Control: **Quinidine Efflux Ratio < 2** (Mean Papp B2A / Mean Papp A2B)
Wells outside of a CV of 50% are discarded from the mean calculation
Slope of the curve **< 1.4 and > 0.8**
Mean recovery > 30% and < 130%. **If recovery is < 30% or > 130% no P_{app} data is reported, only the mean recovery is reported**
Concentration in receiver compartment > LOQ defined by the calibration curve. If all **receiver samples are below LOQ** or not detectable, the **P_{app} quantified at the LOQ will be reported as a < value**. If one or two receiver samples are not detectable or below LOQ, a judgement will be made to exclude these replicates or report the P_{app} quantified at LOQ as a < value
C_0 concentrations must be within +/- 50% of the nominal concentration. Any replicates outside of this range will be excluded from the mean calculation. **If all replicates are outside of this range no results will be reported for Papp, %FA and %recovery.**
If compound failed optimisation, the compound will be reported as **"undefined, low MS signal".**

If a sample tube is empty, it will be reported as **"empty matrix tube".**
All samples must be quantified within the calibration range, any measurements which are above the calibration range will be excluded
% FA: **"< 40"** for a compound with $\mathbf{P_{app}}$ **< 1.5** x10^{-6} cm/s "**40-80**" for a compound with $\mathbf{P_{app}}$ **≥ 1.5** - <10 x10^{-6} cm/s "**80-100**" for a compound with $\mathbf{P_{app}}$ **≥ 10** x10^{-6} cm/s

12. DATA ARCHIVAL

All the raw data, chromatograms and reports should be stored in the server at Aurigene Pharmaceutical Services Limited, India for a period of five years or as specified by sponsor. Assay details and analytical procedures should be captured in LNB and the same should be archived as per the instructions/ procedure of Intellectual Property Management (IPM) department.

13. REFERENCES

- Irvine J. D., Takahashi L. Lockhart K., Cheong J Tola J. W., Selick H. E. and Grove J. R; MDCK CaCO-2 (Madin-Darby Canine Kidney) Cells. A tool for membrane permeability screening. Journal of pharmacology. Science. 88, 28-33; 1999.
- Zhengyin, Gary w. C; Optimization in drug discovery. In Vitro methods. Human Press.19-35; 2004.
- Drug interaction studies – Draft guidelines FDA; 2006.

14. CHANGE AND AMENDMENT HISTORY

Change History		
Version	**Changes made/ Reason for changes**	**Effective date**
2.0	1. Cell line has been changed to MDCK-LE version 2 (MDCK-cMDR1-KO cells) which is provided by Novartis.	20 Apr 2022
	2. Importance of MDCK-LE version 2 cell line in permeability assessment has been included in Introduction (refer section 4)	
	3. Puromycin (2.5 µg/ml) is added extra in the culture media for growing MDCK LE v2 cells (refer section 6.3)	
	4. 0.05 % Trypsin, EDTA, 1x used for trypsinization (Gibco, 25300-054) instead of 10X trypsin-EDTA (Sigma, T4174 or Invitrogen, 15090046; refer section 6.2)	
	5. During trypsinization of cells, an extra step of rising with 1 ml of trypsin before incubation is included. (refer section 7.2)	
	6. Only 5% FBS containing culture media used instead of 10% (refer section 6.3)	
	7. Cells seeding density has been changed from 1,50,000/ml to 50,000 cells/ml (refer section 7.5)	
	8. Transport buffer is replaced with two assay buffers which are 1X HBSS with 0.02 % BSA or 5% BSA, with 1µM Bafilomycin A1 (refer section 6.3), where buffer with 0.02% BSA used for reference, test compound, calibration samples preparation and buffer with 5% BSA used for basal receiver wells	

Change History		
Version	**Changes made/ Reason for changes**	**Effective date**
	during assay (refer section 7.6).	
	9. Sample processing has been changed where there is no 20-fold dilution of C0 and donor samples. Here, all the samples (receiver, donor, C0 and calibration curve samples) are processed in the same manner as mentioned in section 7.7	